UNIVERSITY
OF OSLO

Master’s thesis

The title of my thesis

Any short subtitle

My Name

Mathematics
60 ECTS study points

Department of Mathematics
Faculty of Mathematics and Natural Sciences

Autumn 2025

My Name

The title of my thesis

Any short subtitle

Supervisor:
The Name

Abstract

Here come 3–6 sentences describing your thesis.

This is a relatively short working example of the UiO template for a Master's thesis written in LaTeX. The example is based on Dag Langemyhr's excellent template compliant with the UiO designmanual from 2022, Martin Helsø's instructive examples (2017–2019) and Karoline Moe's seminars on writing in mathematical subjects (2014–2024). We show how some of the standard and most helpful features can be used efficiently, but please note that several of the chapters, sections, packages, tools and environments are optional to include in your thesis and that you must adapt and modify the template to suit your thesis and the traditions in your field of study. If in doubt, ask! Your supervisor and department have the final say in this.

Sammendrag

Her kommer Abstract på et annet språk. Det skal være et sammendrag på norsk i Ph.d.-avhandlinger. Det er like greit å ha det i en masteroppgave også. Det kan være utfordrende å skrive, men det er nyttig.

Dette er en relativt kort tekst der vi bruker UiOs mal for masteroppgaver...

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Preface

The preface is a short and perhaps more personal description of the project, the purpose and your motivation for pursuing it. It should be short and sweet; and it is different from the acknowledgements.

This template has grown out of the work with seminars about scientificness and writing in mathematical subjects that was organized by the Department of Mathematics and the Science Library at the University of Oslo (UiO), and hosted by Karoline Moe, Ph.D., in the period from 2014 to 2024.

There has always been a tradition at the department that former students pass their LaTeX-template on to the next generation of students. However, this practice did not ensure compliance neither with standard traditions in scientific writing and reference management, nor with subsequent design manuals at UiO and typography. Moreover, some of the commands and packages in LaTeX used by students were outdated or in conflict, and there was clearly a need for more instruction.

This template tries to answer these issues. The main purpose of the template is to serve as a starting point for all students using LaTeX to write their Master's thesis. First of all it is a 'minimal' working example with the structure of a thesis. Secondly, We have included examples that most students frequently run into and that can take forever to figure out, even with all available tools, search engines and AI. Finally, we have tried to include a description or short text of the most essential genre-features or pit-falls for each chapter.

We sincerely hope that this template can be of some help in your work, and feel free to send the the LaTeX-group at the University of Oslo Library an e-mail on latexguru@ub.uio.no if you are stuck!

Acknowledgements

chapter:ack

The acknowledgements is where you thank everyone who have contributed to the thesis. It is mandatory to thank your supervisor, co-supervisor, your department and other parties that you have cooperated with; fellow students, a company etc.

The acknowledgements is usually written in a more personal tone than other parts of the thesis, and you should use the pronoun 'I' when you write. F.ex. I would like to thank Dag Langemyhr and Martin Helsø for their passion for LaTeX, their dedication to elegance and detail in typography, and for reading the package documentation at *The Comprehensive TeX Archive Network* (CTAN)[1] so that the rest of us can worry less about layout and more about The Millennium Prize Problems[2].

In the acknowledgements it is custom to include important events that have happened during the period where you were working on the project. Sometimes, this can be done with humor and creativity. But be extremely cautious! Humor changes over time, your life and perspective changes, and you never know who will read this part of your thesis. What seems funny now might be extremely insulting or embarrassing in a couple of years[3].

In fact, the acknowledgements is the part of the thesis that *everyone* who picks up your thesis will read, not just the examiner or the interested researcher. I recommend that you visualize your audience when you feel most creative: your supervisor, your parents, your grandmother, your ex, your future boss, your future co-workers, your future kids, and, perhaps most importantly, your future students.

The fact that your audience might be larger than you think, can not be stressed enough. And this applies to the entire thesis. Your thesis will eventually be deposited in and made openly available to the entire world via the institutional repository DUO. Because of the UiO-address, it gets a high rating on search engines and will be very easy to find for interested researchers all over the world. Moreover, most researchers subscribe to alerts when new citations of their articles or books are published, which means that most of the researchers that you have cited are notified about your thesis, and they might read it, push it to their students, find errors, notice sloppy work and referencing, pick up interesting results and more.

[1] Note that CTAN is the central place for all kinds of material around TeX, with 6649 packages as of October 2024.

[2] Footnotes are usually not used in mathematical texts, in particular not for citations! More on that later. In the meantime, use Oria via ub.uio.no to find books about The Millennium Prize Problems

[3] As a rule of thumb, try to refrain from being too creative and funny in your thesis. Write a blog, diary or send letters to a friend if you must.

0.1 Notation and important results

We will now present some important results that will be used later in the thesis.

thm:bertini **Theorem 0.1.1** (Bertini). *If $\mathcal{D}$ is a linear system on a variety X in characteristic 0, then the general member of $\mathcal{D}$ is smooth away from the base locus of $\mathcal{D}$ and the singular locus of X.*

Add notation (this is from EH): A *linear system* on a scheme X is a pair $\mathcal{D} = (\mathcal{L}, V)$, where $\mathcal{L}$ is a line bundle on X and $V \subseteq H^0(X, \mathcal{L})$ a vector space of sections.

We refer to (!?) for a proof. [EH16 EH16]

Find a good reference for this theorem.

thm:bezout **Theorem 0.1.2** (Bézout). *Let $X, Y \subseteq \mathbb{P}^n$ be two subvarieties having complementary dimension, i.e.* $\dim(X) + \dim(Y) = n$*. If X and Y intersect transversely, then they will intersect in exactly* $\deg(X) \cdot \deg(Y)$ *points.*

We refer to (!?) for a proof. [EH16 EH16] corollary 2.4.

Find a good reference for this theorem.

Definition 0.1.3 ([Ha77 Har77], 12.7). Let Y be a topological space. A function $\phi \colon Y \to \mathbb{Z}$ is *upper semicontinuous* if for each $y \in Y$, there is an open neighborhood U of y, such that for all $y' \in U, \phi(y') \leq \phi(y)$

Lemma 0.1.4 ([EO25 EO25], Cor. 10.59). *Let k be a field, and let $f \colon X \to Y$ be a morphism of projective schemes over k Then f is proper. In particular, the image $f(X) \subseteq Y$ is closed in Y.*

Theorem 0.1.5 (Dimension formula for fibers). *Let k be a field and let $f \colon X \to Y$ be a dominant morphism of irreducible projective varieties over k. Let η denote the generic point of Y and write $X_\eta = X \times_Y \operatorname{Spec} k(Y)$ for the generic fibre. Then*

$$\dim X \;=\; \dim Y \;+\; \dim X_\eta.$$

Proof. For irreducible varieties over a field, the Krull dimension equals the transcendence degree of the function field over k. Dominance of f gives an inclusion of function fields $k(Y) \hookrightarrow k(X)$ and $k(X)$ is a finitely generated field extension of $k(Y)$. Hence

$$\dim X = \operatorname{trdeg}_k k(X) = \operatorname{trdeg}_k k(Y) + \operatorname{trdeg}_{k(Y)} k(X).$$

By definition $\operatorname{trdeg}_{k(Y)} k(X) = \dim X_\eta$, and $\operatorname{trdeg}_k k(Y) = \dim Y$, which yields the claimed equality. □

Write the proof above in own words, and find reference.

Chapter 1

Grassmannians and the space of lines in $\mathbb{P}^3$

assmannians

In this chapter we will introduce Grassmann varieties, or Grassmannians, and focus on $\mathbb{G}(1,3)$, which is the space of lines in $\mathbb{P}^3$. We will also introduce the Plücker embedding, which is a way to embed the Grassmannian into a projective space, giving it the structure of a projective variety. For a more detailed introduction to Grassmannians, we refer to EH16 [EH16].

1.1 Definition and basic properties

A Grassmannian is a projective variety whose closed points correspond to linear subspaces of a certain dimension in an n-dimensional vector space. As a set, $\mathrm{Gr}(k,V)$ is the set of all k-dimensional linear subspaces of the vector space V. In Section 1.2, we will give it the structure of a projective variety. In our case, V will be a vector space over $\mathbb{C}$.

What happens when we use another field than $\mathbb{C}$?. EO works over $\mathbb{Z}$ to get all fields.

A k-dimensional linear subspace of an n-dimensional vector space V can also be thought of as a $(k-1)$-dimensional linear subspace of $\mathbb{P}V \cong \mathbb{P}^{n-1}$. Due to this, we will often write $\mathrm{Gr}(k,V)$ as $\mathbb{G}(k-1,\mathbb{P}V)$. When there is no need to specify the vector space V, only its dimension n, we will write $\mathrm{Gr}(k,n) = \mathbb{G}(k-1,n-1)$.

$$\mathrm{Gr}(k,n)(\mathbb{C}) = \{[L] \mid L \subseteq k^n \text{ is a linear space of dimension } k\}.$$

EO25 [EO25]

1.2 The Plücker embedding

sec:Plucker

1.3 Covering by affine spaces

1.4 Short on Schubert cycles (?)

Decide to include or not.

1.5 Incidence correspondance and the universal Fano scheme

We aim to introduce the powerful concept of incidence correspondences, which will be a tool we will use several times later in this thesis.

EH16
Definition 1.5.1. ([EH16] Section 6.1). N $= \binom{n+d}{d} - 1$

$$\Phi = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(k, n) \mid L \subseteq X\}$$

Proposition 1.5.2. *$\Phi \subset \mathbb{P}^N \times \mathbb{G}(k, n)$ is a closed subset of $\mathbb{P}^N \times \mathbb{G}(k, n)$.*

Decide if we want to prove, or refer to proof. Can refer to Harris 1995 chapter 6, or perhaps proof of 22.31 in EO

Proposition 1.5.3 (EH16, Proposition 6.1). *Let $N = \binom{n+d}{d} - 1$. The universal Fano scheme $\Phi = \Phi(n, d, k) \subseteq \mathbb{P}^N \times \mathbb{G}(k, n)$ is a smooth irreducible variety of dimension*

$$\dim \Phi(n, d, k) = N0(k + 1)(n - k) - \binom{k + d}{k}.$$

Corollary 1.5.4 (EH16, corollary 6.2).

Include result.

EO25
cor:Fano Corollary 1.5.5. *([EO25], 22.31) For a hypersurface $S_d \subseteq \mathbb{P}^3$ of degree d, we expect to find no lines on S_d when $d \geq 4$, finitely many lines when $d = 3$, and positive-dimensional families of lines when $d \leq 2$.*

1.6 Lines in a cubic surface

From Corollary 1.5.5, we know that a smooth cubic surface contains finitely many lines. We will now show that it contains exactly 27 lines.

Theorem 1.6.1. *Each smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines.*

This will be the main theorem of this section. We will prove it in two different ways. The first will be a well known proof, relating cubic surfaces to blow-ups of $\mathbb{P}^2$ in 6 points. The second proof will use intersection theory on the Grassmannian $\mathbb{G}(1,3)$.

Make sure the second result is included in thesis.

We start with the first proof.

1.6.1 Cubic surfaces as blow-ups of $\mathbb{P}^2$ in 6 points

Remark 1.6.2. The points $P_1, \dots, P_6$ in $\mathbb{P}^2$ are said to be in *general position* if no three of the points are collinear, and no six lie on a conic.

Proposition 1.6.3. *(Ha77 [Har77], 4.7/4.7.3) Let $P_1, \dots, P_6 \in \mathbb{P}^2$ are in general position. Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X is isomorphic to the blow-up of $\mathbb{P}^2$ in the points $P_1, \dots, P_6$.*

Proposition 1.6.4. *(Ha77 [Har77], 4.9) Let $X \subset \mathbb{P}^3$ be a smooth cubic surface. Then X contains exactly 27 distinct lines. Each having self-intersection -1, and they are the only irreducible curves with negative self-intersection. They are:*

- *The exceptional curves $E_1, \dots E_6$ (6 of these)*
- *The strict transform of lines in $\mathbb{P}^2$ containing pairs of points P_i, P_j, $i \neq j$ (15 of these).*
- *The strict transform of conics in $\mathbb{P}^2$ not containing P_i (6 of these)*

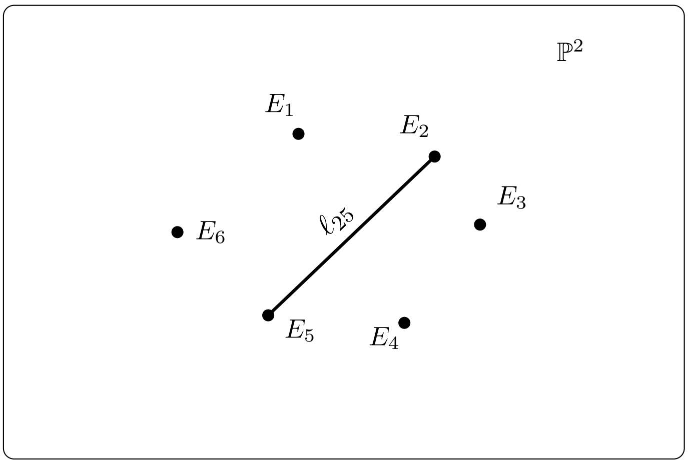


fig:blowup6

Figure 1.1: The projective plane $\mathbb{P}^2$ with six marked blow-up points.

Make a better figure of the blow-up of $\mathbb{P}^2$ in 6 points.

Chapter 2

Results

results

2.1 Surfaces including intersecting lines

In this section we will study the problem of finding a smooth hypersurface $X_d \subset \mathbb{P}^3$ of degree d, containing k lines intersecting at a common point p and lying in a common plane H. We will denote the lines as $l_1, l_2, \ldots, l_k$.

Definition 2.1.1. A k-cross is a union of k lines $l_1, l_2, \ldots, l_k$ in $\mathbb{P}^3$ which intersect at a common point p. We will write $Z_k = \bigcup_{i=1}^k l_i$.

Remark 2.1.2. Note that for automorphisms of $\mathbb{P}^3$, lines are sent to lines. Therefore we can study the lines $l_0 = (\alpha : \beta : 0 : 0),\ l_1 = (\alpha : 0 : \gamma : 0),\ l_2 = (\alpha : \delta : \lambda_2\delta : 0), \ldots, l_k = (\alpha : \delta : \lambda_k\delta : 0)$ for $\lambda_i \in \mathbb{N}$, take the automorphism of $\mathbb{P}^3$ which sends these lines to the given lines, and then study the hypersurface of degree d in $\mathbb{P}^3$ which contains these lines. This automorphism will also takes a smooth hypersurface of degree d to a smooth hypersurface of degree d, so showing the following properties for lines in this form is sufficient for the general case.

We begin with the following lemma of non-existence:

Lemma 2.1.3. *Given k distinct lines $l_1, l_2, \ldots, l_k$ in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane H, there does not exist any smooth hypersurface of degree $2 \leq d \leq k-1$ such that the lines are contained in the hypersurface.*

Proof. Assume for contradiction that there exists a smooth hypersurface of degree $2 \leq d \leq k-1$ such that the lines are contained in the hypersurface. Then we can fix a line l_{k+1} in H, which does not intersect p, and which does not lie in the hypersurface X_d. l_{k+1} will intersect each of the lines at a point p_i, all being distinct since the lines only intersects in p, and thus no other point. l_{k+1} will also intersect the hypersurface at the points p_i, thus intersecting the hypersurface at k points. By Bezout's theorem (Theorem 0.1.2), the intersection of a line with a hypersurface of degree d is at most d points, giving a contradiction. □

We now move on to the case of existence, with the following proposition:

Proposition 2.1.4. *For k lines in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane, we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface.*

Proof. Let $\mathfrak{L}$ be the linear system of hypersurfaces of degree k in $\mathbb{P}^3$ containing the lines $l_0, l_1, \ldots, l_k$. Recall that Bertini's theorem (Theorem 0.1.1) tells us that the general member of $\mathfrak{L}$ is smooth outside the base locus of $\mathfrak{L}$. The base locus of $\mathfrak{L}$ is the union of the lines $l_0, l_1, \ldots, l_k$. Thus we must find a function $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_k$ such that $V(f)$ is smooth on the lines $l_0, l_1, \ldots, l_k$. Bertini's theorem then gives us that we can find a hypersurface which is also smooth outside the base locus of $\mathfrak{L}$.

We study the function $f = g(x_1, x_2) + h(x_0, x_3)$, where $h = x_0^{k-1}x_3$ and $g = \prod l_i = x_1x_2\prod_{i=3}^k (x_1 - \lambda_i x_2)$, giving the following polynomial:

$$f(x_0, x_1, x_2, x_3) = x_1x_2 \prod_{i=3}^{k} (x_1 - \lambda_i x_2) + x_0^{k-1}x_3$$

We can see that f is a polynomial of degree k, and that f includes the lines $l_0, l_1, \ldots, l_k$, since $f(\alpha : \beta : 0 : 0) = f(\alpha : 0 : \gamma : 0) = f(\alpha : \delta : \lambda_i\delta : 0) = 0$. We must then check that f is smooth on the base locus of $\mathfrak{L}$, which is to check that $\bigcap_{i=0}^3 (\partial_i f = 0) = \emptyset$.

We get the following derivatives:

$$\begin{aligned}
\partial_0 f &= \partial_0 g + \partial_0 h = 0 + (k-1)x_0^{k-2}x_3 = (k-1)x_0^{k-2}x_3 \\
\partial_1 f &= \partial_1 g + \partial_1 h = x_2 \prod_{i=3}^{k} (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^{k} \prod_{\substack{j=3 \\ j\neq i}}^{k} (x_1 - \lambda_j x_2) \\
\partial_2 f &= \partial_2 g + \partial_2 h = x_1 \prod_{i=3}^{k} (x_1 - \lambda_i x_2) + x_1x_2 \sum_{i=3}^{k} (-\lambda_i) \cdot \prod_{\substack{j=3 \\ j\neq i}}^{k} (x_1 - \lambda_j x_2) \\
\partial_3 f &= \partial_3 g + \partial_3 h = 0 + x_0^{k-1} = x_0^{k-1}
\end{aligned}$$

We then find that

- $\partial_0 f = 0$ on the entire base locus.
- $\partial_1 f = 0$ on the line $l_0 = (\alpha : \beta : 0 : 0)$.
- $\partial_2 f = 0$ on the line $l_1 = (\alpha : 0 : \gamma : 0)$.
- $\partial_3 f = 0$ if and only if $x_0 = 0$, which is the case for the following points: $(0 : \beta : 0 : 0), (0 : 0 : \gamma : 0), (0 : \delta : \lambda_i\delta : 0)$ for $3 \leq i \leq k$.

We can see that the intersection of all these zero locuses is empty, showing that the function f is smooth on the base locus of $\mathfrak{L}$. Thus we can find a smooth hypersurface of degree k such that the lines are contained in the hypersurface.

□

By the results above, we get the following corollary:

Corollary 2.1.5. *Given k lines in $\mathbb{P}^3$ intersecting at a common point p and lying in a common plane, we can find a smooth hypersurface of degree d such that the lines are contained in the hypersurface if and only if $d \geq k$ or $d = 1$.*

Proof. If we have k lines, we can find a smooth hypersurface of degree $d \geq k+1$ by including arbitrary lines $l_{k+1}, l_{k+2}, \ldots, l_d$ lying in the plane. For the case $d = 1$, we can take the plane H containing the lines. □

2.2 The set of degree d surfaces including k-crosses

It is natural to ask about the set of degree d surfaces including k-crosses. Let Z_k be a k-cross lying in a plane H, which we will assume is the plane $\mathbb{P}^2_{x,y}$ by an automorphism of $\mathbb{P}^3$.

Proposition 2.2.1. *Let $A_k = \{S_d \subseteq \mathbb{P}^3 \mid Z_k \subset S_d\} \subseteq \{S_d \subseteq \mathbb{P}^3\}$ be the set of degree d surfaces containing Z_k. Then $\dim A_k = \binom{d+1}{2}d - k + 1$.*

Proof. $S_d = V(f)$ for some $f \in \mathbb{C}[x_0, x_1, x_2, x_3]_d$. Furthermore, since Z_k is included in the x, y plane, we write f as $f = g(x, y) + zh(x, y, z)$, seperating the part of f that vanishes on the plane. Since $Z_k \subseteq S_d$, we have that $g(x, y)$ vanishes on Z_k. That $C_k \subseteq S_d$ is the same as saying that $f \mid_{C_k} = 0$, which is the same as saying that $g(x, y) \mid_{C_k} = 0$. This is because $zh(x, y, z) \mid_{C_k} = 0$ since $C_k \subseteq \mathbb{P}^2_{x,y} = V(z)$. This is implies that $l_1 l_2 \cdots l_k \mid g(x, y)$, meaning that $g(x, y) = l_1 l_2 \ldots l_k \cdot j(x, y)$ for some $j(x, y) \in \mathbb{C}[x, y]_{d-k}$. Thus we can write f as $f = l_1 l_2 \cdots l_k \cdot j(x, y) + zh(x, y, z)$, where $h(x, y, z) \in \mathbb{C}[x, y, z]_{d-1}$.

Finishing the argument, we get that $\dim A_k = \dim\langle j(x, y), h(x, y, z)\rangle = \binom{d-k+1}{1} + \binom{d+1}{2}$. □

2.3 Surfaces including k-gons

In this section, we will consider k-gons in $\mathbb{P}^3$ and their relationship with hypersurfaces.

Definition 2.3.1. A k-gon is a union of k lines $l_1, l_2, \ldots, l_k$ such that l_i intersects l_{i+1} for $1 \leq i < k$, and l_k intersects l_1. We will write $C_k = \bigcup_{i=1}^k l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < k$, and $p_k = l_k \cap l_1$.

Proposition 2.3.2. *Let $C_k = \bigcup_{i=1}^k l_i$ be a k-gon in $\mathbb{P}^3$. Then we can find a hypersurface of degree d such that $C_k \subseteq S_d$ when $k \leq \frac{(d+3)(d+2)(d+1)}{6d}$.*

Proof. We can look at the zero cohomology of the ideal sheaf $\mathcal{I}_{C_k}$ of the k-gon to see if the k-gon can be contained in a hypersurface. We will denote the union of the lines as $C_k = \bigcup_{i=1}^k l_i$.

We have the short exact sequence

$$0 \to \mathcal{I}_{C_k} \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_{C_k} \to 0.$$

We twist this by $\mathcal{O}_{\mathbb{P}^3}(d)$ to get

$$0 \to \mathcal{I}_{C_k}(d) \to \mathcal{O}_{\mathbb{P}^3}(d) \to \mathcal{O}_{C_k}(d) \to 0.$$

From this we get the long exact sequence in cohomology:

$$\begin{tikzcd}
0 \arrow[r] & H^0(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) \arrow[r] & H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^0(C_k, \mathcal{O}_{C_k}(d)) \arrow[dll, rounded corners] & \\
& H^1(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) \arrow[r] & H^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^1(C_k, \mathcal{O}_{C_k}(d)) \arrow[dll, rounded corners] & \\
& H^2(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) \arrow[r] & H^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^2(C_k, \mathcal{O}_{C_k}(d)) \arrow[dll, rounded corners] & \\
& H^3(\mathbb{P}^3, \mathcal{I}_{C_k}(d)) \arrow[r] & H^3(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(d)) \arrow[r] & H^3(C_k, \mathcal{O}_{C_k}(d)) \arrow[r] & 0
\end{tikzcd}$$

We know that $H^i(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = 0$ for $i > 0$. This means we can focus on the first part of the sequence:

$$0 \to H^0(\mathbb{P}^3, \mathcal{I}_{C_k}) \to H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) \to H^0(C_k, \mathcal{O}_{C_k}) \to H^1(\mathbb{P}^3, \mathcal{I}_{C_k}) \to 0.$$

Since h^0 is additive, we get

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_k}) = h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) + h^1(\mathbb{P}^3, \mathcal{I}_{C_k}) - h^0(C_k, \mathcal{O}_{C_k}).$$

By Thm 18.4 in (EO) we get that $h^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) = \binom{d+3}{3}$. $h^0(C_k, \mathcal{O}_{C_k}(d))$ can be computed by considering the short exact sequence:

$$0 \to \mathcal{O}_{C_k}(d) \to \bigoplus_{i=1}^{k} \mathcal{O}_{l_i}(d) \to \bigoplus_{i=1}^{n} \mathcal{O}_{p_i} \to 0.$$

This gives us $h^0(C_k, \mathcal{O}_{C_k}) = k \cdot \binom{d+1}{1} - k = kd$.

$$h^0(\mathbb{P}^3, \mathcal{I}_{C_k}) \geq \binom{d+3}{3} - kd.$$

Meaning we can guarantee a degree d surface when $\binom{d+3}{3} - kd \geq 0$, or equivalently when $k \leq \frac{(d+3)(d+2)(d+1)}{6d}$.

$\square$

To get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_k})$, we would need to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_k})$. This issue will be explored in section (!?).

Remark 2.3.3. Note that this argument does not guarantee that the surface is smooth, only that it exists.

2.4 Computational results

Improve this section.

We have done computational experiments in Macaulay2 to get a better understanding of the inequality above. We have tested the existence and smoothness of hypersurfaces including k-gons for various degrees d and number of lines k. The code used for these experiments is included in the following snippet. We will discuss the details of the code below, along with its results.

lst:k-gon

Listing 2.1: k-gon

```
-- Step 1: Define the ring and parameters
kk = ZZ/32749;
kk = QQ;  -- Uncomment to work over the rationals
RingP3 = kk[x_0..x_3];
d = 2;  -- Degree of the hypersurface
nrPoints = 4;

-- Step 2 and 3: Generate random points and create pairs of subsequent points
-- Function for generating i random points in P^3
randomPoints = i -> (
    v := for j in 0..3 list random(kk);
```

```
    -- Ensure the point is not zero
    while all(v, x -> x == 0) do (
        v = for j in 0..3 list random(kk)
    );
    v
)
-- Pick nrPoints random points in P^3
myPoints = apply(nrPoints, randomPoints);
-- Create pairs of subsequent points, and also the pair consisting of the last and first point
subsequentPairs = for i from 0 to (#myPoints - 2) list {myPoints#i, myPoints#(i+1)};
-- Add final pair
subsequentPairs = append(subsequentPairs, {myPoints#0, myPoints#(#myPoints-1)});

-- Step 3: Generate matrices from point pairs
-- Function to convert two points to a matrix with first row {x_0 .. x_3}
pointsToMatrix = points -> matrix {{x_0 .. x_3}, points#0, points#1};
-- Convert each pair of points to a matrix
pointsMatrix = apply(subsequentPairs, pointsToMatrix);

-- Step 4: Get the ideals of the lines
-- Given a matrix with first row {x_0 .. x_3} and next two rows points, return the ideal of the line through the points
lineFromPoints = pointsMatrix -> minors(3, pointsMatrix);
allLines = apply(pointsMatrix, lineFromPoints);

-- Step 5: Create the ideal of the union of all lines
-- Intersect the ideals of the lines to get ideal of the union of all lines
Iz = intersect allLines;

-- Step 6: Pick a degree d element in Iz, which defines a surface containing all the lines, and check properties
-- Pick a random degree d element in Iz
f = random(d, Iz);
-- Create the variety defined by f
Vf = variety ideal f;

-- Check if the surface is smooth and not the zero polynomial
codimension = codim singularLocus ideal f;
isZero = (f == 0)
isSingular = (codimension < 4)
isSmooth(Vf) and not isZero
```

We'll go through the code step by step.

Step 1: We begin by defining the field and the polynomial ring in four variables over this field. We then define the degree d of the surfaces we want to consider, and the number of points to use, which will be the same as the number of lines in the k-gon.

Step 2: We define a function to pick a given amount of random points in our quotient ring, ensuring they are not zero. We then apply the function to get the points, and gather the points into pairs of subsequent points so that we can later define lines between these.

Step 3: Next, we define a function for turning two points into a matrix. The function takes two points and returns a matrix, as follows:

$$((a,b,c,d),(d,e,f,g)) \mapsto \begin{pmatrix} x_0 & x_1 & x_2 & x_3 \\ a & b & c & d \\ d & e & f & g \end{pmatrix}$$

Step 4: Now we can define the ideal of the line between the two points by taking the 3×3 minors of this matrix.

Step 5: Next we take the intersection of all the line ideals, to get the ideal of $Z = \bigcup l_i$,

Step 6: We choose a random member of $I_Z(d)$ and compute if this defines an empty, singular, or smooth surface.

Results: We have been interested in finding the maximum number of lines k such that we can find a smooth surface of degree d including a k-gon. The results are summarized in the following table:

2.5 Cohomology of a 4-gon

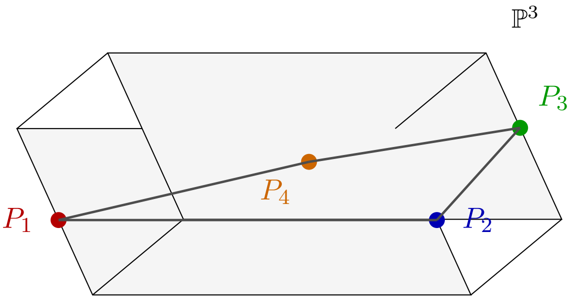
As mentioned earlier, to get an exact value for $h^0(\mathbb{P}^3, \mathcal{I}_{C_k})$, we would need to compute $h^1(\mathbb{P}^3, \mathcal{I}_{C_k})$. We will now explore this issue for the case of a 4-gon and looking at a suitable resolution of the ideal sheaf of the 4-gon. First we claim that the intersection points of a 4-gon can, by an automorphism of $\mathbb{P}^3$, be expressed as the points $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$. Then the 4-gon can be expressed as the union of the lines $l_1 = V(x_2,x_3), l_2 = V(x_0,x_3), l_3 = V(x_0,x_1), l_4 = V(x_1,x_2)$. Now the ideal of the 4-gon is given by $I_{C_4} = (x_2,x_3) \cap (x_0,x_3) \cap (x_0,x_1) \cap (x_1,x_2) = (x_1x_3, x_0x_2)$. We justify our chose of points in the following lemma.

Lemma 2.5.1. *Let C_4 be a 4-gon in $\mathbb{P}^3$. Then there exists an automorphism of $\mathbb{P}^3$ such that the intersection points of C_4 are given by the points* $(1:0:0:0), (0:1:0:0), (0:0:1:0), (0:0:0:1)$.

Proof. Let p_1, p_2, p_3, p_4 be the intersection points of C_4. To be in general position in $\mathbb{P}^3$, we require that no three of the points are collinear, and that the points do not all lie in a plane. We can then find a projective transformation sending p_1 to $(1:0:0:0)$, p_2 to $(0:1:0:0)$, and p_3 to $(0:0:1:0)$. Since no three of the points are collinear, p_4 cannot be sent to a point on the line between any two of the other points. Furthermore, since the points do not all lie in a plane, p_4 cannot be sent to a point in the plane spanned by any three of the other points. Thus, we can send p_4 to $(0:0:0:1)$. $\square$

We will now look at a resolution of the ideal sheaf of the 4-gon. We have that $I_{C_4} = (x_1x_3, x_0x_2)$. This ideal has the following free resolution:

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-4) \xrightarrow{\begin{pmatrix} -x_0x_2 \\ x_1x_3 \end{pmatrix}} \mathcal{O}_{\mathbb{P}^3}(-2)^{\oplus 2} \xrightarrow{\begin{pmatrix} x_1x_3 & x_0x_2 \end{pmatrix}} I_{C_4} \to 0.$$



$P_1 = [1:0:0:0]$
$P_2 = [0:1:0:0]$
$P_3 = [0:0:1:0]$
$P_4 = [0:0:0:1]$

Make a better figure of the 4-gon in $\mathbb{P}^3$.

Finish argument.

2.6 k-chains

Definition 2.6.1. A k-chain is a union of k lines $l_1, l_2, \ldots, l_k$ such that l_i intersects l_{i+1} for $1 \leq i < k$. We will write $C_k = \bigcup_{i=1}^{k} l_i$ and denote the intersection points as $p_i = l_i \cap l_{i+1}$ for $1 \leq i < k$.

Notice that a k-chain is a k-gon without the line l_k intersecting l_1. One can then modify the code above to create k-chains instead of k-gons, by excluding the final pair of points. The results from this are similar to those of k-gons, in that for higher degrees, we find smooth surfaces, up until we find no surfaces. There are however interesting exceptions for degree 2 and 3. Here we get that for 4 lines and 6 lines, we get singular surfaces for 4 and 7 lines, respectively. We will handle the cases seperately.

Proposition 2.6.2. *For a general 4-chain in $\mathbb{P}^3$, we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Proof. First, assume that S_2 is smooth. Since S_2 is a smooth quadric surface in $\mathbb{P}^3$, it is of the form $S_2 = \mathbb{P}^1 \times \mathbb{P}^1$. This means that S_2 has two rulings, meaning that there are two families of lines on S_2, such that any two lines in the same family do not intersect, and any two lines in different families intersect at exactly one point. Denote the two families as F_1 and F_2. Then we can see that a 4-chain must consist of two lines from F_1 and two lines from F_2, since any two lines in the same family do not intersect. This means that the 4-chain must be of the form l_1, l_2, l_3, l_4 where $l_1, l_3 \in F_1$ and $l_2, l_4 \in F_2$. However, this means that l_4 intersects l_1, since they are in different families, meaning that the 4-chain is in fact a 4-gon. This is a contradiction, since we assumed that we had a general 4-chain. Thus we cannot find a smooth quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain. □

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Proposition 2.6.3. *For a general 4-chain in $\mathbb{P}^3$, we can find a singular quadric surface $S_2 \subset \mathbb{P}^3$ such that it includes the 4-chain.*

Proof. Find a nice example of a singular quadric surface including a 4-chain. □

Definition 2.6.4. We define $Q_m \subseteq \mathbb{G}(1,3)^m$ to be the subset of m-tuples of lines $(l_1, \ldots, l_m)$ such that they form an m-chain.

lem:dimQm

Lemma 2.6.5. *Q_m is an irreducible variety of dimension $3m+1$ for $m \geq 1$.*

verify

Irreducibility is dicussed in EH 3.3.1, saying schubert cells and their closures are irreducible.

Is $m \geq 1$ necessary?

Proof. We observe that Q_m can be constructed as a product of Grassmannians, with a condition of intersection added for $m \geq 2$. That is, if we are given a line $L \in \mathbb{G}(1,3)$, we can consider the space of lines intersecting L, denoted as $\Sigma_L = \{l \in \mathbb{G}(1,3) \mid l \cap L \neq \emptyset\}$. Then $Q_m = \mathbb{G}(1,3) \times \Sigma_L^{m-1}$. For m = 1, we have that $Q_1 = \mathbb{G}(1,3)$, which is irreducible of dimension 4. Now, assume that the statement holds for $m-1$. Consider the projection map $\pi : Q_m \to Q_{m-1}$ given by $\pi(l_1, \ldots, l_m) = (l_1, \ldots, l_{m-1})$. The fiber over a point is given by $\pi^{-1}(l_1, \ldots, l_{m-1}) = \{l \in \mathbb{G}(1,3) \mid l \cap l_{m-1} \neq \emptyset\}$.

Should we denote the fiber over a point as the whole m-tuple, or just the last line?

Note that for a given line L, Σ_L can be constructed as the image of the incidence correspondence $\Gamma = \{(l,p) \in \mathbb{G}(1,3) \times L \mid p \in l\}$ to $\mathbb{G}(1,3)$. The fiber of the projection map $f : \Gamma \to L$ is given by the set of lines through a point $p \in L$, which is isomorphic to $\mathbb{P}^2$. Then the fiber of each point in L is irreducible of dimension 2, and since f is proper, we have that Γ is irreducible of dimension 3, meaning that Σ_L is irreducible of dimension 3. Thus, Q_m is the product of $m-1$ irreducible varieties of dimension 3, and one Grassmannian of dimension 4, so Q_m is irreducible of dimension $4+3(m-1) = 3m+1$. This completes the proof. □

Finish proof. It seems this can be explained by schubert cycles, 3.3.1 in EH. This can maybe be a subsection in preliminaries. The proof can be modified if this is included in preliminaries.

Now we will look at the the case of 7-chains in cubic surfaces.

Proposition 2.6.6. *For a general 7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain.*

Proof. Let $\mathcal{M}$ be the space of smooth cubic surfaces in $\mathbb{P}^3$. This is an open subset of $\mathbb{P}(\mathbb{C}[x_0, \ldots, x_3]_3) \cong \mathbb{P}^{19}$. Since $\mathcal{M}$ is an open subset, it shares the dimension, i.e $\dim \mathcal{M} = 19$. Let $\mathcal{N} = \{(S_d, l_1, \ldots, l_7) \in \mathcal{M} \times \mathbb{G}(1,3)^7 \mid \mathcal{L} \subset S_d, \mathcal{L} \text{ is a } k\text{-chain}\}$ be the incidence variety of smooth cubic surfaces including a 7-chain. We have the following diagram:

Why open?

$$
\begin{tikzcd}
 & \mathcal{N} \subseteq \mathcal{M} \times \mathbb{G}(1,3)^7 \arrow[dl, "\pi_1"'] \arrow[dr, "\pi_2"] & \\
\mathcal{M} & & \mathbb{G}(1,3)^7
\end{tikzcd}
\tag{2.1}
$$

Should we use Q_7 instead of $\mathbb{G}(1,3)^7$?

The projection map $\pi_1 : \mathcal{N} \to \mathcal{M}$ is given by $\pi_1(S_d, l_1, \ldots, l_7) = S_d$. Then the fiber over a point $S_d \in \mathcal{M}$ is given by the set of 7-chains included in S_d. Since a smooth cubic surface includes exactly 27 lines, we can find a 7-chain by choosing 7 lines from these 27. We know from (!?) that it is possible to find a 7-chain, and so this fiber is not empty. Furthermore, there are only a finite amount 7-chains when there are a finite amount of lines. Then the dimension of the fiber $\dim pi_1^{-1}(S_d)$ is 0, being a finite set of 7-chains. Thus, from (!?), we have that $\dim \mathcal{N} = \dim \mathcal{M} + \dim \pi_1^{-1}(S_d) = 19 + 0 = 19$.

Refer to result

Result in this section

Refer to theorem about dimension of fiber.

Now we are interested in the fiber of the set of 7-chains in $\mathbb{P}^3$, $Q_7 \subseteq \mathbb{G}(1,3)^7$. By lemma lem:dimQm 2.6.5, we have that $\dim Q_7 = 22$, and so the projection map $\pi_2 : \mathcal{N} \to \mathbb{G}(1,3)^7$ given by $\pi_2(S_d, l_1, \ldots, l_7) = (l_1, \ldots, l_7)$ cannot be surjective on Q_7. This means that taking a general 7-chain $C_7 \in Q_7$, and considering the fiber $\pi_2^{-1}(C_7)$, we do not expect this to be non-empty, meaning we will not have a surface including it. Thus, for a general

7-chain in $\mathbb{P}^3$, we cannot find a smooth cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. ☐

Proof should include: projection is surjective (every smooth cubic surface has at least a 7-chain, shown in this section), implies dim N $\geq$ dim M. Then fiber is finite, so dim N $\leq$ dim M. Prop 10.26?

Again, as in the case for 4-chains in quadrics, computations show that we can find a singular cubic surface including a 7-chain.

Proposition 2.6.7. *For a general 7-chain in $\mathbb{P}^3$, we can find a singular cubic surface $S_3 \subset \mathbb{P}^3$ such that it includes the 7-chain. (!?)*

Proof. Find a nice example of a singular cubic surface including a 7-chain. ☐

Note that if we instead asked whether a smooth cubic surface in $\mathbb{P}^3$ can include a 7-chain, the situation is different. In this case, we can indeed find a 7-chain such that it is included in the surface. In fact, we know we could find a 12-chain in this case, and a 13-gon. We gather show this in the following proposition.

Move this proposition earlier, to use in proof. Maybe lemma? Also interesting in itself, so maybe not.

Proposition 2.6.8. *For a smooth cubic surface in $\mathbb{P}^3$, we can find a 13-gon such that it is included in the surface.*

Proof. A smooth cubic surface in $\mathbb{P}^3$ is isomorphic to the blow-up of $\mathbb{P}^2$ in 6 points (!?). The surface then includes 27 lines, which are precisely the exceptional divisors of the blow-up, the strict transforms of the 15 lines through pairs of points, and the strict transforms of the 6 conics through 5 of the points. We can then find a 13-gon by considering the following lines: Let $E_1, E_2, E_3, E_4, E_5, E_6$ be the exceptional divisors, let L_{ij} be the strict transform of the line through points p_i and p_j, and let C_i be the strict transform of the conic through all points except p_i. Then we can consider the 13-gon given by the lines $E_1, L_{12}, E_2, L_{23}, E_3, L_{34}, E_4, L_{45}, E_5, L_{56}, E_6, C_1, L_{16}$. This is indeed a 13-gon since each line intersects exactly two other lines in the set. ☐

By removing one of the lines from the 13-gon, we immediately get the following:

Corollary 2.6.9. *For a smooth cubic surface in $\mathbb{P}^3$, we can find a 12-chain such that it is included in the surface.*

This implies that the k-chains longer than 7 included in the smooth cubic surfaces are somehow special.

Fill out later.

Remark 2.6.10. (!?) Just as we can find for which n and d we expect a general hypersurface $X \subseteq \mathbb{P}^n$ of degree d to contain lines by looking at the universal Fano scheme, we can do the same for k-chains. $N = \binom{n+d}{d} - 1$. $\mathcal{L} = \bigcup_{i=1}^{k} l_i$. Instead of looking at $\Phi = \{(X, L) \in \mathbb{P}^N \times \mathbb{G}(1,3) \mid L \subset X\}$, we instead observe $\mathcal{N} = \{(S_d, l_1, \ldots, l_7) \in \mathbb{P}^N \times \mathbb{G}(1,3)^7 \mid \mathcal{L} \subset S_d, \mathcal{L} \text{ is a } k\text{-chain}\}$. Let $\mathcal{M} = \{S_d \subseteq \mathbb{P}^n \mid S_d \text{ smooth}\}$.

Fix G(k,n). k is overloaded.

2.7 Spørsmål til møte

- Hvordan vise at kravet om snitt gir dimensjon 3? Bruke Schubert celler? Bruke dim av fibre (dim bilde = dim L = 1, dim fiber = dim P2 = 2)?
- Dobbeltsjekk at idealet viser at det kun er én singulær flate som inneholder en 7-kjede.
- Kilde for insidens korrespondanse/variete.
- Eksempler på ting jeg er usikker på om jeg bør bevise. Hvordan best skrive at referanse til hvor det bevises?
- Skal jeg refere til alle bevis som jeg ikke viser selv?

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